

ITD224 Project – Data Science**Student ID:** 4061218C**Name:** Tan Hui Kiang (Judy)**Class:** ITSPA9/2026S1**Lecturer:** Mr Adrian Eng**Date:** 24 Aug 2026**Submission:** Final Presentation and Final Project Portfolio (Individual)

HDB Resale Price Analytics and Prediction

Data Cleaning, Exploratory Data Analysis, Modelling and Evaluation

Project outcome: A tuned Histogram Gradient Boosting model achieved MAE S\$37,274.69, RMSE S\$53,549.33 and R^2 0.937 on the chronologically held-out 2026 test set.

Executive Summary

This individual project applies the CRISP-DM approach to Singapore HDB resale transactions from January 2017 to July 2026. The business problem is that resale-price estimation is complex because prices depend on location, floor area, flat type, lease profile, storey and changing market conditions. A data-driven model can provide an objective reference price to support homeowners, buyers and property analysts.

The raw dataset contained 236,059 transactions and 11 variables. Data preparation removed 316 exact duplicates, standardised formats, converted remaining lease into numeric years, created time and storey features, and retained legitimate premium transactions while flagging potential outliers. The final modelling dataset contained 235,743 records and 24 variables, with no missing essential values.

Three regression algorithms were compared using chronological validation: Ridge Regression, Random Forest and Histogram Gradient Boosting. Histogram Gradient Boosting was selected and tuned because it gave the best validation performance with practical training time. On the final 2026 test set, the tuned model achieved MAE S\$37,274.69, RMSE S\$53,549.33 and R^2 0.937. These results show strong explanatory and predictive performance, but the model should be used as a decision-support range rather than a formal valuation.

Recommended use: Use the prediction as a pricing benchmark, display an uncertainty range of at least $\pm$ MAE, and require further review for premium towns, unusual flat types and high-value outliers.

Assessment Coverage

Assessment requirement	Evidence in this report
1. Business problem, dataset and cleaning Quick Recap	Sections 1-3
2. Model design - Select Modelling Technique - Generate Test Design - Build Model	Sections 4-5
3. Model Assessment - Determine model assessment criteria - Assess Model	Section 6
4. Evaluation and recommendations - Evaluate model and review the model's construction if it properly achieves the business objectives - Identify business issues that should be addressed earlier - Make relevant recommendations	Sections 7-8

1. Business Goal and Objectives

1.1 Business Problem

HDB resale prices vary substantially across towns, flat types and time periods. A price based only on floor area or recent nearby transactions may overlook lease duration, storey, property age, flat model and broader market movements. The project therefore addresses the need for a consistent and explainable resale-price benchmark.

1.2 Business Goal

To develop a reliable regression model that predicts HDB resale prices from transaction and flat characteristics, enabling evidence-based pricing decisions and reducing reliance on subjective judgement.

1.3 Project Objectives

- Prepare a clean, consistent and modelling-ready HDB resale transaction dataset.
- Explore the factors associated with resale price and translate them into business insights.
- Engineer meaningful time, lease, age and storey variables.
- Compare at least three regression techniques using MAE, RMSE and R^2 .
- Tune the strongest model and evaluate it on an unseen chronological test set.
- Interpret feature importance, identify limitations and recommend responsible business use.

1.4 Success Criteria

- The selected model should materially outperform the linear baseline.
- R^2 should be sufficiently high to explain most observed price variation.

- MAE and RMSE should be practical relative to typical transaction values.
- Performance should be reviewed by town and flat type, not only as an overall average.
- Recommendations must acknowledge uncertainty and avoid presenting predictions as guaranteed valuations.

2. Dataset Description

The project uses transaction-level HDB resale data covering January 2017 to July 2026. Each row represents a registered resale transaction. The original dataset contains 236,059 records and 11 variables.

Variable	Type	Purpose
month	Date	Transaction timing and market trend
town	Categorical	Location effect
flat_type	Categorical	Size/configuration category
block, street_name	Categorical	Address identifiers; excluded from baseline modelling
storey_range	Categorical	Converted into storey midpoint
floor_area_sqm	Numeric	Physical size
flat_model	Categorical	Design/model differences
lease_commence_date	Numeric	Lease-age reference
remaining_lease	Text	Converted into numeric years
resale_price	Numeric	Target variable

2.1 Data Scope and Assumptions

- The dataset represents registered transactions, not asking prices or formal valuations.
- The 2026 period is partial-year data through July and is used only as the final test period.
- Macroeconomic variables, distance to MRT, school proximity, renovation quality and exact coordinates are not included.
- Historical relationships may change after policy, interest-rate or market-regime shifts.

3. Data Preparation and Transformation

3.1 Acquire and Select Data

The transaction dataset was loaded into pandas and copied before transformation to preserve the raw data. The target variable was resale_price. Predictors were selected based on relevance, availability before prediction and the risk of redundancy or excessive cardinality.

3.2 Clean Data

1. Standardised column names and cleaned leading/trailing spaces in text fields.
2. Removed 316 exact duplicate transactions, reducing the dataset from 236,059 to 235,743 rows.
3. Converted month to a proper date and numeric fields to appropriate data types.
4. Checked essential modelling variables; no rows were removed for missing or invalid essential values.
5. Applied validity rules for positive floor area, lease, storey and resale price.

3.3 Construct Data

Engineered variable	Construction	Business meaning
remaining_lease_years	Years + months/12	Remaining ownership tenure
resale_year/month/quarter	Extracted from transaction month	Market timing and seasonality
flat_age_at_resale	Resale year - lease commencement	Age/depreciation proxy
storey_low/high/mid	Parsed from storey range	Floor-level premium
price_per_sqm	Resale price / floor area	EDA benchmark only; excluded from predictive features to avoid target leakage
outlier flags	IQR-based indicators	Identify unusual but potentially genuine transactions

3.4 Integrate and Format Data

Text categories were standardised to uppercase, dates were stored consistently, and modelling variables were retained in a single cleaned dataset. High-cardinality address fields and the raw remaining-lease text were excluded from the baseline model. The cleaned file contained 235,743 records and 24 variables.

3.5 Outlier Treatment

The IQR method identified 4,331 resale-price outliers (1.84%), 1,257 floor-area outliers (0.53%) and 9,015 price-per-square-metre outliers (3.82%). These observations were flagged rather than automatically deleted because premium locations, large executive flats and uncommon units can be valid transactions. This preserves real market diversity while enabling targeted error analysis.

4. Exploratory Data Analysis

EDA was performed to understand market activity, price distribution and relationships between resale price and potential predictors. The analysis used bar charts, trend plots, boxplots, distributions, scatterplots and correlation heatmaps.

Finding	Evidence	Modelling/business implication
Floor area is influential	Correlation with price = 0.565	Retain as a core numeric predictor
Prices increased over time	Average: S\$443,743 in 2017 to S\$660,087 in partial 2026	Include transaction timing
Location matters	Bukit Timah had the highest historical average at S\$787,624	Encode town as categorical
Lease and age matter	Remaining lease positive; flat age negative	Represent depreciation, but monitor redundancy
Higher storeys receive a premium	Storey midpoint correlation = 0.34	Retain storey midpoint
Price is right-skewed	Premium-value tail and valid outliers	Use robust tree-based models and segment review

4.1 Correlation Interpretation

Among modelling-relevant numeric variables, floor area had the strongest positive correlation with resale price (0.56), followed by resale year (0.41), storey midpoint (0.34) and remaining lease (0.30). Correlation measures linear association only and does not capture nonlinear interactions or categorical location effects. Price per square metre had a stronger correlation but was excluded from modelling because it is calculated using the target and would cause data leakage.

EDA conclusion: Resale price is driven by a combination of property size, location, lease profile, storey and time. No single variable is sufficient, supporting the use of multivariate machine-learning regression.

5. Model Design

5.1 Data-Mining Objective

The analytical task is supervised regression: predict a continuous target, resale_price, from characteristics available at the time of estimation. The model is intended for decision support, comparative pricing and market analysis.

5.2 Predictor Selection

Categorical predictors were town, flat type and flat model. Numeric predictors were floor area, resale year, resale month, flat age at resale, remaining lease years and storey midpoint. Address identifiers were excluded to control high cardinality, while price per square metre was excluded to prevent target leakage.

5.3 Selected Modelling Techniques

Model	Rationale	Expected trade-off
Ridge Regression	Transparent regularised linear baseline	Fast but limited for nonlinear interactions
Random Forest	Captures nonlinearities and interactions	Slower and may generalise less well over time

Model	Rationale	Expected trade-off
Histogram Gradient Boosting	Efficient boosting for large tabular data	Requires tuning and is less directly interpretable

5.4 Preprocessing Pipelines

For Ridge Regression, missing values were imputed, numeric variables were standardised and categories were one-hot encoded. For tree-based models, categories were ordinal-encoded with handling for unseen values. All preprocessing was placed inside scikit-learn pipelines to ensure consistent treatment and reduce leakage risk.

5.5 Test Design

Partition	Period	Records	Purpose
Training	2017-2024	196,686	Fit initial models
Validation	2025	25,072	Compare models and tune parameters
Final test	Jan-Jul 2026	13,985	One-time evaluation on unseen future data

A chronological split was selected instead of a random split because the intended use is to estimate future transactions. This design provides a more realistic test of temporal generalisation and prevents future records from influencing model selection.

6. Model Assessment

6.1 Assessment Criteria

- MAE: average absolute prediction error in Singapore dollars; easy to communicate.
- RMSE: gives greater weight to large errors; important for premium or unusual transactions.
- R^2 : proportion of variation in resale prices explained by the model.
- Segment MAE and median absolute percentage error: checks whether overall results hide weaker towns or flat types.
- Training time and model complexity: practical considerations for reproducibility and future updating.

6.2 Initial Validation Results

Model	MAE (S\$)	RMSE (S\$)	R^2	Training (s)
Histogram Gradient Boosting	50,942.05	69,119.16	0.885	6.89
Random Forest	51,564.48	73,481.21	0.870	83.74
Ridge Regression	62,142.20	87,644.04	0.816	1.23

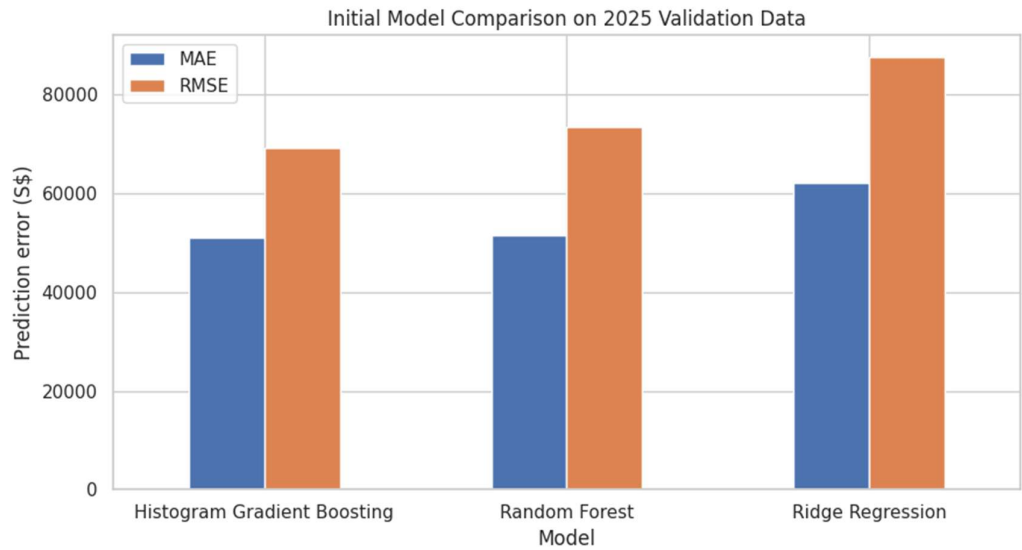


Figure 1. Initial model comparison on the 2025 validation set.

Histogram Gradient Boosting provided the lowest validation MAE and RMSE and the highest R^2 . It also trained much faster than Random Forest. It was therefore selected for hyperparameter tuning.

6.3 Hyperparameter Tuning

Six transparent candidate configurations were evaluated on the 2025 validation set. Candidate 6 produced the best validation RMSE of S\$65,461.76 and R^2 of 0.897 using `learning_rate=0.04`, `max_iter=500`, `max_leaf_nodes=127`, `min_samples_leaf=30` and `l2_regularization=5.0`.

6.4 Final Test Performance

Metric	Final result	Interpretation
MAE	S\$37,274.69	Average absolute difference between prediction and actual price
RMSE	S\$53,549.33	Larger errors remain important, particularly in premium segments
R^2	0.937	Model explains 93.7% of test-set price variation
Median APE	4.35%	Half of test predictions were within about 4.35% absolute error



Figure 2. Actual versus predicted resale prices on the 2026 final test set.

7. Evaluation Against the Business Objective

7.1 Does the Model Achieve the Objective?

The model substantially outperformed the Ridge baseline and achieved R^2 of 0.937 on future-period data. This indicates that the selected variables and modelling approach capture most systematic price variation. An average error of approximately S\$37,275 is useful as an initial pricing benchmark, especially when paired with comparable transactions and an uncertainty range.

However, business success is not defined by R^2 alone. The model's RMSE is higher than MAE, showing that some large errors remain. The residual mean of S\$5,851 indicates slight overall underprediction on the 2026 test period. Performance also differs across segments: Central Area recorded MAE of about S\$73,061, while Bukit Batok recorded about S\$23,860. Multi-Generation flats had only four test records and should not be treated as a stable segment result.

7.2 Feature Importance

Permutation importance showed that floor area and town were the dominant drivers, followed by flat type, remaining lease, flat age and storey midpoint. This is consistent with the EDA and with real-estate pricing logic. Resale year had zero permutation importance within the 2026-only test set because it was constant across the test period; this does not mean time is unimportant across the full historical dataset.

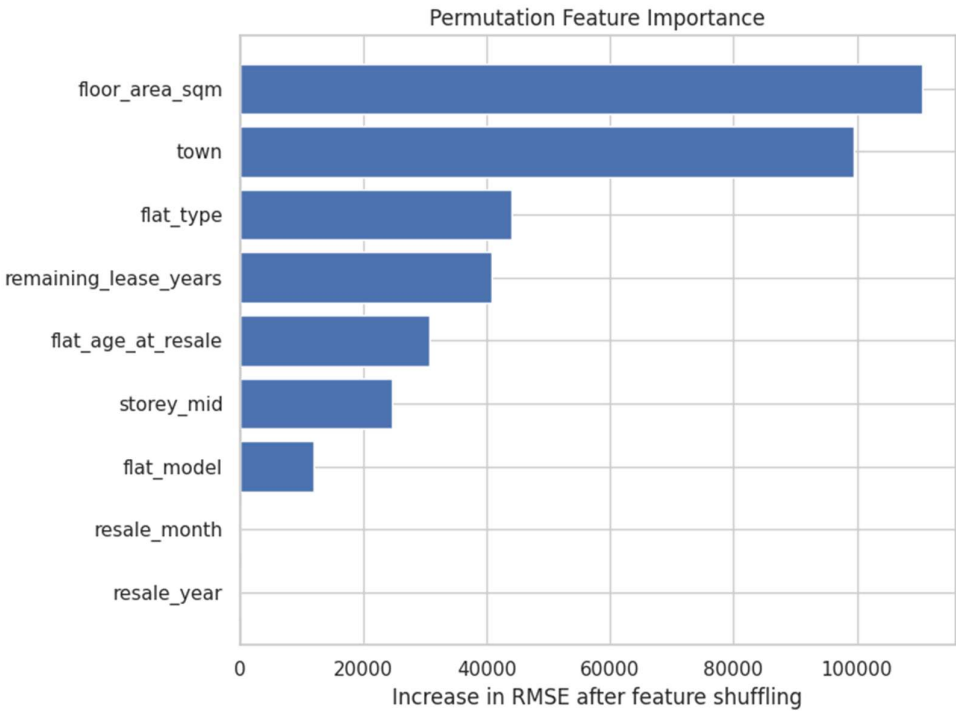


Figure 3. Permutation feature importance on a 5,000-record test sample.

7.3 Business Issues That Should Have Been Addressed Earlier

- Geospatial detail: town is broad; exact coordinates, MRT distance, schools and amenities could improve local accuracy.
- Policy and macroeconomic context: interest rates, grants, cooling measures and supply changes may shift market relationships.
- Renovation and unit condition: transaction data does not record internal condition, view, orientation or renovation quality.
- Definition of acceptable error: stakeholders should agree whether tolerance is absolute, percentage-based or segment-specific.
- Premium and rare segments: explicit minimum sample sizes and escalation rules should be defined before deployment.
- Partial-year test period: the 2026 result should be revalidated when a complete year becomes available.

8. Recommendations

Priority	Recommendation	Business value
1	Present prediction with a range of at least ±\$37,275 and comparable transactions.	Communicates uncertainty and discourages false precision
2	Flag premium towns, rare flat types and predictions outside historical ranges for manual review.	Reduces risk from weak or sparse segments
3	Add geospatial and amenity variables such as MRT distance, schools and town-centre proximity.	Improves local price differentiation
4	Retrain periodically and monitor MAE, RMSE, bias and segment performance.	Detects market drift and policy changes

Priority	Recommendation	Business value
5	Use explainability outputs and retain an audit trail of model version, inputs and prediction date.	Supports transparency and governance
6	Benchmark against a simple comparable-sales method before operational use.	Confirms incremental business value

Appendix A.

Project Proposal & Milestone Review – Submitted on 09 Aug 2026

Appendix B.

Final Project Presentation – completed on 24 Aug 2026 via Zoom

Appendix C.

GitHub portfolio: [https://github.com/judytanhk-sin/4061218C_ITD224_Final_Project_Portfolio]

Notebook filename: 4061218C_ITD224_Final_Notebook.ipynb